

Introduction and Overview of k -means clustering

Albert Y. Kim

Defining “Machine Learning”

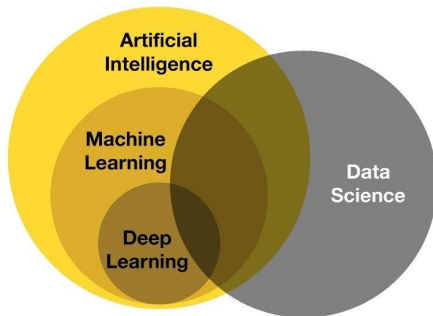
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- ▶ **Supervised learning:**

- ▶ **Classification** - how to classify each sample into a discrete category
- ▶ **Regression** - how to predict a numeric quantity corresponding to each sample

- ▶ **Unsupervised learning:**

- ▶ **Clustering** - how to categorize the *samples* into meaningful groups (that aren't predefined)
- ▶ **Dimension reduction** - how to meaningfully represent each sample using $p^* < p$ features by finding patterns among the original features
- ▶ **Anomaly detection** - how to identify samples that deviate significantly from “normal data”

First steps

- ▶ We'll devote most of our attention this semester to supervised learning, but we'll start with a short unit on unsupervised learning for a few reasons:
 - ▶ It's manageable to learn Python, unsupervised learning algorithms, and notation simultaneously
 - ▶ Unsupervised methods can be used as *pre-processing steps* to prepare data for supervised learning

Notation

To begin, let's consider **tabular data** that is organized in a matrix, **X**, consisting of n samples (rows) with p features (columns)

$$\mathbf{X} = \begin{pmatrix} x_{1,1} & x_{1,2} & \cdots & x_{1,p} \\ x_{2,1} & x_{2,2} & \cdots & x_{2,p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n,1} & x_{n,2} & \cdots & x_{n,p} \end{pmatrix}$$

In this class I'll try to stick to the following conventions:

1. Bold capital letters to denote a matrix
2. Bold lower-case letters to denote a vector
3. non-bold capital letters to denote a random variable
4. non-bold lower-case letters to denote a scalar value

Clustering

- ▶ The goal of clustering is categorize our n samples into groups
 - ▶ A simple approach is to find k non-overlapping groups, where k is pre-specified
 - ▶ As such, the first approach we'll discuss is **k-means clustering**

k-means clustering (set up)

- ▶ Consider a set of **prototypes** defined by the following coordinates in the p -dimensional space of our data:

$$\{\mathbf{z}_1, \dots, \mathbf{z}_k\}$$

- ▶ Let $A()$ denote an *assignment function* that assigns a sample to the group associated with a prototype
 - ▶ For example, $A(\mathbf{x}_i) = j$ indicates the i^{th} sample is assigned to the j^{th} prototype

k-means clustering (objective function)

- ▶ Generally speaking, machine learning methods occurs by optimizing an **objective function** (in supervise learning we'll call this a *cost function*)
 - ▶ In *k*-means clustering, we seek to minimize the following:

$$\sum_{i=1}^n \|\mathbf{x}_i - \mathbf{z}_{A(\mathbf{x}_i)}\|^2 = \sum_{i=1}^n (\mathbf{x}_i - \mathbf{z}_{A(\mathbf{x}_i)})^T (\mathbf{x}_i - \mathbf{z}_{A(\mathbf{x}_i)})$$

- ▶ The data (the $\mathbf{x}_i$'s) are fixed, and we can simplify things by assuming an assignment function, such as the nearest prototype by euclidean distance: $d(\mathbf{x}_1, \mathbf{x}_2) = \sqrt{\sum_{j=1}^k (x_{1,j} - x_{2,j})^2}$
 - ▶ *k*-means clustering amounts to solving for the prototype coordinates that minimize this objective function

k-means (algorithm)

The *k*-means algorithm alternates between two steps:

1. Given the current prototype coordinates, assign each sample to the nearest prototype using euclidean distance (ie: apply $A()$ to each sample).
2. Update the prototype coordinates to now be the mean coordinates of their respective assigned samples.

These steps repeat until additional iterations no longer improve the objective function.

k -means (algorithm demo)

Here is animation similar to what we walked through on the board:

- ▶ [Link to \$k\$ -means GIF animation](#)

k-means (initialization)

Unfortunately, *k*-means is sensitive to the initial set of prototype coordinates. The `sklearn` implementation allows us two options:

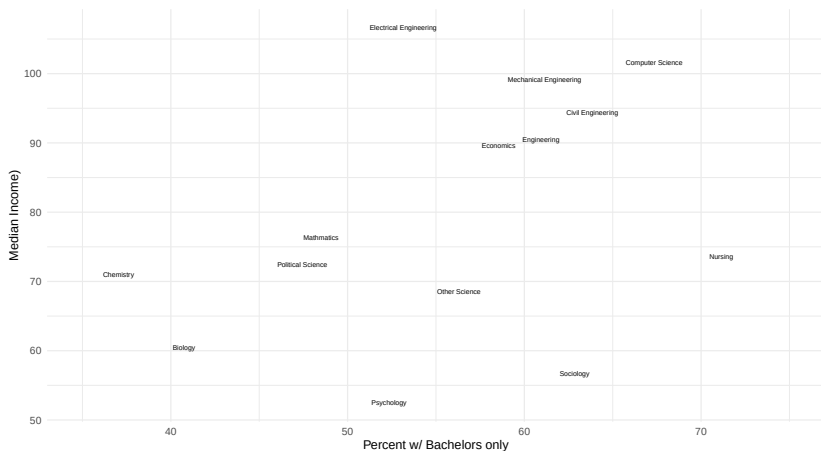
1. `random` (also known as Forgy) - randomly select *k* of the observed samples and use their coordinates as the initial prototypes
2. `k-means++` - choose the first prototype uniformly at random from the observed samples, then use probability distributions find additional prototypes that are far from existing prototypes ([click here for details](#))

k -means (the role of k)

- ▶ The k -means algorithm requires a pre-specified value of a k , making it an example of what's known as a **hyperparameter** or **tuning parameter**
- ▶ A simple way to intelligently decide upon the value of hyperparameter is a method known as **grid search**, which is an exhaustive search over an enumerated set of candidate values
 - ▶ In the context of k -means, we might look at the value of the objective function over a sequence of k values
 - ▶ We can then visually display this relationship and select k at an inflection point (where additional prototypes no longer produce the same improvement in the object function)

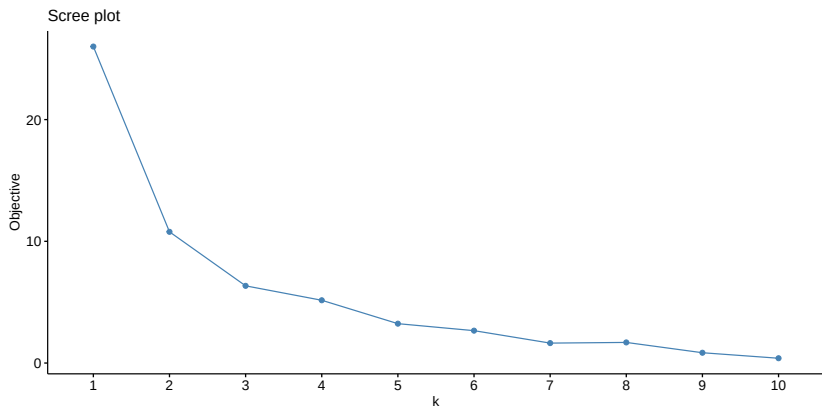
k-means (example - starting data)

These are data from Census's ACS income by degree field survey:



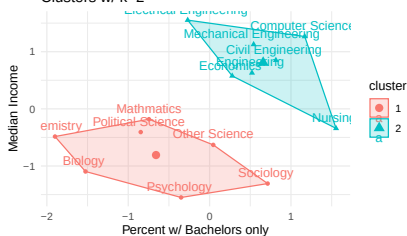
k -means (example - objective vs. k)

Below is the value of the objective function for several choices of k , notice how it levels off:

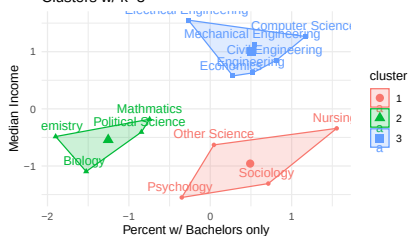


k-means (example - clustering results)

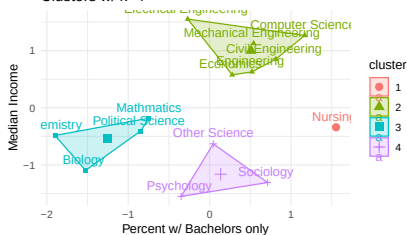
Clusters w/ k=2



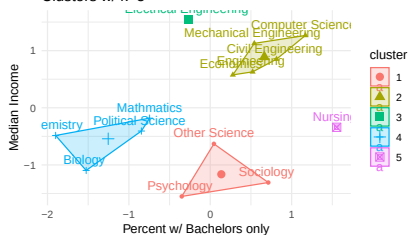
Clusters w/ k=3



Clusters w/ k=4



Clusters w/ k=5



Conclusion

- ▶ *Supervised learning* aims to predict a numeric outcome or classify data-points after learning the patterns present in the available data
- ▶ *Unsupervised learning* finds patterns in the available data without specifying a target outcome
 - ▶ *k*-means clustering is an example of an unsupervised learning method
- ▶ Today's lab will be longer than normal and will cover both the basics of Python as well as how to utilize *k*-means clustering in Python with an application to image segmentation